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B. Tech. (First & Second Semester) Examination, Nov.-Dec. 2020

(New AICTE Scheme)

(AEI, Biotech, Chem., Civil, CSE, Elect., EEE, EI, Et & T, IT, Mech., Mining, Metallurgy, Mechatronics, Production, Automobile Agriculture Branch)

PHYSICS-I

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt **five** questions branchwise. Part (a) of each question is compulsory and carries 4 marks each. Attempt any **two** parts from (b), (c) and (d) of each question carrying 8 marks.*

5 Units branchwise :

I. Civil/Metallurgy/Mining— Units 1/2/3/8/10

II. Mechanical/Mechatronics/Production/Automobile—Units 1/4/5/6/10

III. Electrical/Electrical and Electronics/Chemical Engg.—Units 1/3/7/8/9

IV. Computer Science/IT/Electronics/EI/AEI/Biotech—Units 1/7/8/9/10

Useful Physical Constants :

(i) Charge of electron 'e' = $1{\cdot}6\times10^{-19}$ C

(ii) Mass of electron 'm_e' = $9{\cdot}0\times10^{-31}$ kg

(iii) Speed of light 'c' = 3×10^{8} m/s

(iv) Plank's constant 'h' = $6{\cdot}62\times10^{-34}$ J-s

(v) Boltzmann constant 'k' = $1{\cdot}38\times10^{-23}$ J/K

(vi) Mass of proton 'm_p' = $1{\cdot}6725\times10^{-27}$ kg

(vii) Avogadro number 'N_A' = $6{\cdot}023\times10^{26}$/kg-mol

(viii) Permittivity of a vacuum 'ϵ_0' = $8{\cdot}85\times10^{-12}$ F/m

(ix) 1 eV = $1{\cdot}602\times10^{-19}$ J

(x) 1 a.m.u. = 931 MeV = $1{\cdot}66\times10^{-27}$ kg

Unit-I

1. (a) Match List-I (units) and List-II (Physical quantity)

and select the correct answer using the codes given below the lists :

(Give explanation for correct answer also)

List-I (units)	List-II (Physical quantity)
A Watt	1 Electric charge
B Tesla	2 Power
C Coulomb	3 Luminous intensity
D Candela	4 Magnetic field

(i) A-2, B-4, C-1, D-3

(ii) A-1, B-2, C-3, D-4

(iii) A-1, B-2, C-4, D-3

(iv) A-2, B-4, C-3, D-1 4

(b) Derive expressions for :

(i) Time of flight

(ii) Equation of trajectory and

(iii) Maximum height attained

If the projectile rises above the point of projection. 8

(c) Describe the relativity velocity in detail. Obtain an expression for the cases when two bodies are moving along a straight line in the same direction and opposite direction. 8

(d) (i) If velocity, time and force were choosen the basic quantities, find the dimension of mass. 3

(ii) A man walks on a straight road from his residence to the office 2·5 km away with a speed of 5 km/hour. Finding the office closed, he immediately turn and walks back to his residence with a speed of 7·5 km/hour. What is the magnitude of average velocity and average speed of the man over the interval of time : (a) 0 to 30 min, (b) 0 to 50 min and (c) 0 to 40 min. 5

Unit-II

2. (a) What is parallel axis theorem and to whom it is applied? (Give explanation also)

Choose the correct answer :

(i) Theorem used to add the two mutually perpendicular moment of inertia for areas.

(ii) Theorem used to add the two mutually perpendicular moment of inertias for volumes.

(iii) Theorem used to add the two mutually perpendicular moment of inertia for linear distances.

(iv) Theorem used to add the two mutually perpendicular moment of inertia for vectors. 4

(b) Obtain an expression for the moment of inertia of a body about a given line on its surface using the parallel axis theorem. 8

(c) Write short notes on any **two** of the followings :

(i) Bending stress and shear stress

(ii) Concepts of elasticity and plasticity

(iii) Moment of inertia calculations

(iv) Angular velocity and accelerations 8

(d) (i) The motor of an engine is rotating about its axis with an angular velocity of 100 rev/minute. It comes to rest in 15 second, after being switched off. Assuming constant angular deceleration. Calculate the number of revolutions made by it before coming to rest. 4

(ii) A wheel of perimeter 220 cm rad/s. On a level road at a speed of 9 km/h. How many revolutions does the wheel make per second. 4

Unit-III 28

3. (a) The minimum thickness of a film which will strongly reflect the light of wavelength 589 nm, if the refractive index of the material of the film is 1·25, will be :

(Choose the correct answer and give explanation also) :

(i) 1·18 mm

(ii) 118 nm

(iii) 128 nm

(iv) 1·18 Å 4

(b) Explain how interference fringes may obtained with the help of a Fresnel's Bi-prism. How may be the arrangement be used to determine the wavelength of a given monochromatic source of light? 8

(c) What is meant by resolving power of a diffraction grating? Show that the resolving power is given by the product of the total number of ruling and the order of spectrum. 8

(d) (i) Newton's rings formed by sodium light between a flat glass plate and a convex lens are viewed normally, what will be the order

of the dark ring which will have double the diameter of that of 40th dark ring? 4

(ii) A square piece of cellophone film with index of refractive 1·5 has a wedge shaped section so that its thickness of the two opposite sides is t_1 and t_2. If with a light of $\lambda =$ 6000 Å, the number of fringes appearing on the film is 10, calculate the difference $(t_2 - t_1)$. 4

Unit-IV 28

4. (a) Identify the nature of the field if the divergence is zero and curl is also zero.

(Choose the correct answer and explain the answer also) :

(i) Solenoidal, irrotational

(ii) Divergent irrotational

(iii) Solenoidal, rotational

(iv) Divergent rotational 4

(b) Explain the electrostatic potential for a charge distribution. Calculate the electric field and

electrostatic potential for a charge distribution in a line segment. 8

(c) What is dielectric constant? How the permeability is related with dielectric constant of the medium. 8

(d) Consider two concentric sphere of radii a and b, $a < b$. The outer sphere is kept at a potential v_0 and the inner sphere at zero potential. Solve Laplace equation in spherical co-ordinates, to find :

(i) The potential and electric field in the region between two spheres.

(ii) Find the capacitance between them. 8

Unit-V 28

5. (a) Given the vector potential is 16-12 sin yj. Find the field intensity at the origin.

(Choose the correct answer and give proper explanation for it) :

(i) 28

(ii) 16

(iii) 12

(iv) 4 4

(b) What do you mean by divergence and curl of a vector field? State and prove Stoke's theorem, related to curl of a vector field. 8

(c) Calculate the magnetic potential due to a bar magnet. Discuss effect of position of the point. Derive an expression for magnetic intensity at any point due to a bar magnet. 8

(d) (i) A wire placed along north-south direction carries a current of 10A from south to north. Find the magnetic field due to 1 cm piece of wire at a point 200 cm north-coast from the piece. 4

(ii) An iron ring of mean diameter of 100 cm and 10 cm^2 cross section has 1000 turns of insulated copper wire wound uniformly on it. If it produces a flux of $2{\cdot}5\times10^{-3}$ webers, when a current of 8 amperes flows in it, calculate the relative permeability of the medium. 4

Unit-VI 28

6. (a) The amplitude of the electric field in a parallel beam of light of intensity $2{\cdot}0\ Wm^{-2}$ will be :

(Choose the correct answer with explanations) :

(i) $3{\cdot}2\times10^{-18}$ NC^{-1}

(ii) 38 NC^{-1}

(iii) $3{\cdot}8\times10^{-12}$ NC^{-1}

(iv) $38{\cdot}8$ NC^{-1} 4

(b) Show that vector E and H are always in phase with each other and at right angles with each other and the wave propagates in $(E\times H)$ direction for an electromagnetic wave travelling in free space. 8

(c) By using Maxwell's equations develop wave equation for a transverse electric and magnetic fields in free space. Also derive an expression for the velocity of electromagnetic waves. 8

(d) (i) A copper rod AB is moving with a velocity of 5m/s parallel to a long straight wire carrying a current of 10 A. Calculate induced e.m.f. in the rod if its lower end A is b (= 18 cm) away from wire and its upper end B is a (= 2 cm) away from the wire as shown in the figure given below : 5

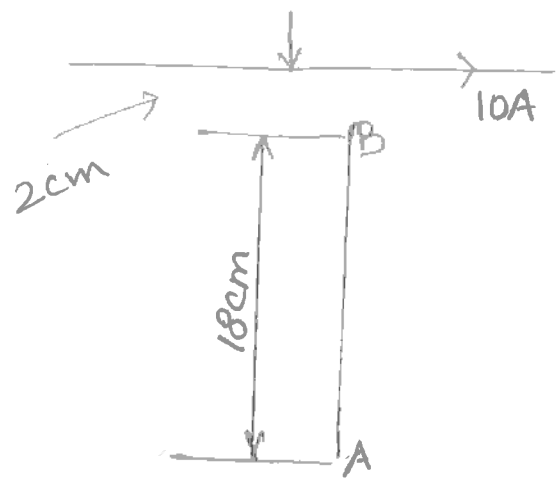

Fig. 1

(ii) A 12 V battery connected to a 6Ω, 10 H coil through a switch drives a constant current in the circuit. The switch is suddenly opened. Assuming that it took 1 ms to open the switch, calculate the average emf induced across the coil. 3

Unit-VII 28

7. (a) An electron has a speed of 600 m/s with an accuracy of 0·005%, what will be the certainty with which we can locate the position of the electron.

(Choose the correct answer with proper explanation) :

(i) 0·0022 m

(ii) 22·002 m

(iii) 0·02254 m

(iv) 22·54 m 4

(b) Write down Schroedinger equation for a particle in a box. Solve it to obtain eigen functions and show that the eigen values are discrete. 8

(c) Derive Schroedinger time dependent equations for matter waves. What is the physical significance of ψ used in this equation? 8

(d) (i) Assume that the uncertainty in the location of a particle is equal to its de-Broglie wavelength. Show that the uncertainty in the velocity is equal to its velocity. 4

(ii) Find the function, $f(x)$ for which x,

$f(x) = -\frac{i}{h} a^2 P_x f(x)$ where a is the real quantity? 4

`Unit-VIII 28

8. (a) The energy of a photon of sodium light $(\lambda = 589\ \text{nm})$ equals the band gap of a semiconducting material. What will be the minimum energy E required to create hole-electron pair :

(Choose the correct option and explain it) :

(i) 2·2 eV

(ii) 2·002 eV

(iii) 22·2 eV

(iv) 2·1 eV 4

(b) What do you mean by density of states and energy band diagram? Deduce an expression for the density of energy states. 8

(c) Discuss the Kronig-Penny model for the motion of an electron in a periodic potential. Explore the conclusions regarding E-K diagrams in solids. 8

(d) Calculate the free electron concentration mobility and drift velocity of electron in aluminium wire of length of 5 mtr and resistance $0 \cdot 06\,\Omega$ carrying a current of 15A. Assuming that each aluminium atom contributes 3 free-electrons for conduction.

Given that :

Resistivity of aluminium = $2{\cdot}7\times10^{-8}\ \Omega$-m

Atomic weight = 26·98

Density = $2{\cdot}7\times10^{3}$ kg/m^2

Avagadro number = $6{\cdot}025\times10^{23}$ 8

Unit-IX 28

9. (a) What will be the diffusion coefficient of electron in silicon at 300 K if electron mobility (μ_e) is 0·19 m^2/v-s :

(Choose the correct answer and explain it) :

(i) $4{\cdot}18\times10^{-6}$ m^{2}s^{-1}

(ii) $94{\cdot}81\times10^{-4}$ m^{2}s^{-1}

(iii) $49{\cdot}18\times10^{-4}$ m^{2}s^{-1}

(iv) $4{\cdot}91\times10^{-4}$ m^{2}s^{-1} 4

(b) What is an intrinsic semiconductors? Write the expression for electron and hole concentration and deduce from them that the fermi-energy level is at the middle of the conduction band and valence band of the intrinsic semiconductors? 8

(c) Write short notes on any two of the following :

(i) Diffusion and drift current

(ii) Charge neutrality condition

(iii) Concept of fermi level

(iv) Depletion region and potential barrier 8

(d) (i) The following data are given for intrinsic germanium at 300 K.

$n_i = 2{\cdot}4\times10^{19}/\text{m}^3$

$\mu_e = 0{\cdot}39\ \text{m}^2\ \text{v}^{-1}\ \text{s}^{-1}$

$\mu_p = 0{\cdot}19\ \text{m}^2\ \text{v}^{-1}\ \text{s}^{-1}$

Calculate the resistivity of the sample. 4

(ii) for the silicon semiconductor with band gap 1·12 eV, determine the position of the fermi level at 300 K, if $m_e^* = 0{\cdot}12\ \text{m}_0$ and $m_n^* = 0{\cdot}28\ \text{m}_0$. 4

Unit-X

10. (a) A signal of 100 MW is injected into a fiber. The outcoming signal from the other end is 40 MW. What will be the loss in dB?

(Correct answer with explanation should be choosen from the following options) :

(i) 9·38 dB

(ii) 8·93 dB

(iii) 3·00 dB

(iv) 3·98 dB 4

(b) Explain the purpose of an active medium in a gas laser. With the help of suitable diagram, explain the working principle, constructional details and uses of a Helium-Neon laser. 8

(c) What is angle of acceptance? Describe the mechanism of propagation of signal through an optical fiber and prove that :

$$\theta_{i(max)} = \sin^{-1} \sqrt{\frac{\mu_1^2 = \mu_2^2}{\mu_0}}$$

Where μ_1, μ_2 are the refractive indices of core and cladding respectively. 8

(d) (i) Find the relative population of the two states in a Ruby laser that produces a light beam of wavelength 6943Å at 300 K. 4

(ii) The numerical aperture of an optical fiber is 0·39. If the differences in the refractive indices of the material of its core and the cladding is 0·05. Calculate the refractive indices of the material of the core. 4